Lab Task

Part 1: House Price Prediction Scenarios

Scenario 1: Predicting House Prices from Size

Question:

You are a data analyst working for a real estate agency. The agency wants to help clients estimate the selling price of their homes based on a simple feature: the square footage of the house. You have been given a small dataset of recently sold houses, containing their size in square feet and their final selling price.

Your task is to:

- Build a simple linear regression model that predicts the house price based on its size.
- Use the model to predict the price of a house that is 2,100 square feet.
- Explain the relationship (the model's slope) between the size of a house and its price.

How to Do:

Model Building: We will use the scikit-learn library in Python to create a linear regression model. The 'Size' of the house will be our independent variable (X), and the 'Price' will be the dependent variable (Y). The model will learn a linear relationship of the form Price = (Slope * Size) + Intercept.

Prediction: After training the model on the provided data, we will input 2100 into our model to get the estimated price.

Relationship Explanation: The slope of the regression line tells us how much the price is expected to increase for each one-unit (one square foot) increase in the size of the house. A positive slope means that as the size of the house increases, its price tends to increase as well. The intercept represents the baseline price of a house, even if it had zero square footage (which is a theoretical value but necessary for the model's equation).

Scenario 2: Predicting House Prices from Multiple Features

Question:

Your real estate agency wants a more sophisticated model. They believe that both the size of the house and the number of bedrooms significantly influence the price. You are given an updated dataset that now includes the

number of bedrooms.

Your task is to:

- Build a multiple linear regression model that predicts house price based on both size and the number of bedrooms.
- Predict the price of a house that is 2,100 square feet and has 4 bedrooms.
- Evaluate how well the model fits the data using the R-squared value and explain what it means.

How to Do:

Model Building: We will build a multiple linear regression model, which extends simple linear regression by considering more than one independent variable. The model will take the form: Price = (Slope1 * Size) + (Slope2 * Bedrooms) + Intercept.

Price Prediction: We will provide the new house's features (2100 for size, 4 for bedrooms) to the trained model to get a price estimate.

Model Evaluation (R-squared): The R-squared value (Coefficient of Determination) tells us the proportion of the variance in the house prices that is predictable from the size and number of bedrooms. An R-squared value closer to 1 indicates that the model explains a large portion of the variability in prices and is a good fit for the data.

Part 2: Iris Dataset Regression Scenarios

Scenario 3: Predicting Petal Length from Petal Width

Question:

A botanist has observed a strong correlation between the petal width and petal length of Iris flowers. They want to create a model to quantify this relationship and predict the length of a petal if they only know its width.

Your task is to:

- Build a simple linear regression model using the Iris dataset to predict petal length (dependent variable) based on petal width (independent variable).
- Use the trained model to predict the petal length for a flower with a petal width of 1.5 cm.
- Visualize the relationship with a scatter plot and the regression line.

How to Do:

Model Building: We will use the 'petal width (cm)' column from the Iris dataset as our independent variable (X) and the 'petal length (cm)' column as our dependent variable (Y). A simple linear regression model will be trained on this data.

Prediction: We will input a petal width of 1.5 cm into the model to get the predicted petal length.

Visualization: A scatter plot of all the data points (petal width vs. petal length) will be created. We will then overlay the linear regression line on this plot. This visualization will clearly show the linear relationship the model has learned and how well it fits the data.

Scenario 4: Predicting Sepal Length from Sepal Width

Question:

Following the success of the petal model, the botanist now wants to see if a similar predictive relationship exists between the sepal measurements. They are skeptical that sepal width is a good predictor for sepal length.

Your task is to:

- Build a simple linear regression model to predict sepal length (dependent variable) based on sepal width (independent variable).
- Evaluate the model's performance using the R-squared value.
- Based on the R-squared value, conclude whether sepal width is a good predictor for sepal length.

How to Do:

Model Creation: We will create a linear regression model where 'sepal width (cm)' is the independent variable (X) and 'sepal length (cm)' is the dependent variable (Y).

Model Evaluation: After training, we will calculate the R-squared value. This metric will tell us what percentage of the variability in sepal length can be explained by the sepal width.

Conclusion: A low R-squared value (e.g., close to 0) would indicate that there is a weak linear relationship between sepal width and sepal length, and the model does not fit the data well. This would support the botanist's skepticism. A high R-squared value would suggest a stronger relationship. By looking at the plot and the R-squared score, we can make a data-driven conclusion.